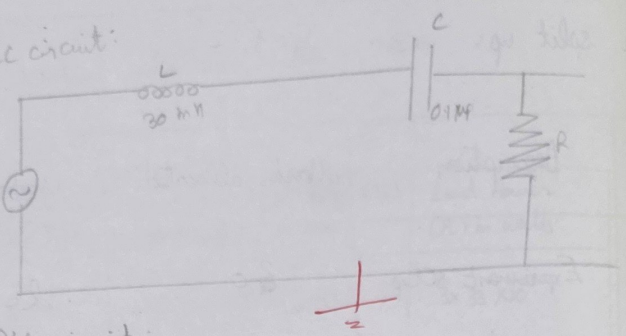
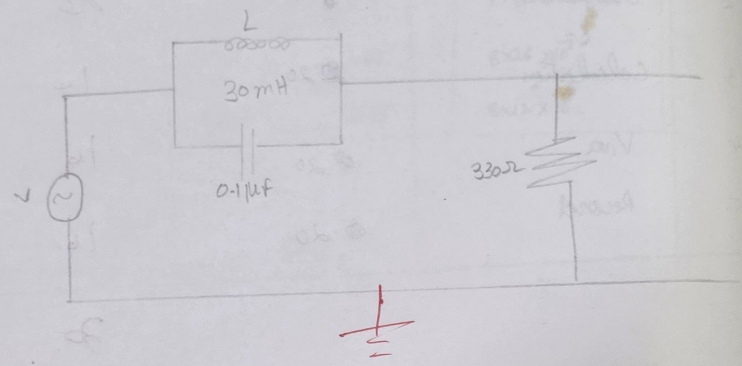
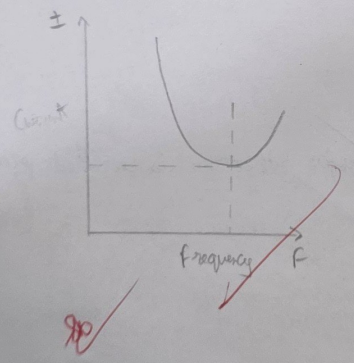
Series RLC circuit:



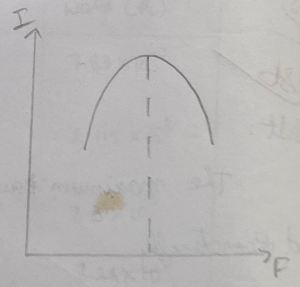
Parallel RLC circuit:



Parallel resonance:



Series resonance:



Ex.No: 7
10/3/26

FREQUENCY RESPONSE OF SERIES AND PARALLEL RESONANCE CIRCUIT

Aim:

To study the behaviour of series and parallel RLC circuit at resonance and to determine the resonance frequency, bandwidth and Q of the given RLC circuit using multisim simulator

Apparatus required:

S No	Component	Specification	Qty
1.	CRO	10 MHz	1
2.	B/B	–	1
3.	Capacitor	0.1 μf	1
4.	Resistor	330 Ω	1
5.	Bread board	–	–
6.	Connectin wire	–	–
7.	PC multisim simulator	–	–

Tabulation $V_{in} = 4V_{p-p}$

SERIES RESONANCE			PARALLEL RESONANCE		
Frequency (Hz)	Output voltage (V)	Output current (mA)	Frequency (Hz)	Output voltage (V)	Output current (mA)
500	1.2	2	500	1.5	9
700	2.0	4	700	2.8	7
900	2.5	7	900	4.0	5
1000	5.0	10	1000	5.0	2
1100	3.6	7	1100	4.1	5

Calculation:

$L = 100\,mH = 0.1\,H$

$C = 0.25\,\mu F = 0.25 \times 10^{-6}\,F$

$$f_0 = \frac{1}{2\pi\sqrt{(0.1)(0.25\times10^{-6})}}$$

$$= \frac{1}{2\pi\sqrt{1.38\times10^{-4}}}$$

$$= \frac{1}{9.92\times10^{-4}} = 1000\,Hz$$

Maximum current (series) and maximum current (parallel) occurs at 1000 Hz

$$Q_w = R/2\pi L = \frac{330}{2\times3.14\times0.1} = 525.4$$

$$Q = \frac{f_0}{B_w} = \frac{1000}{525.4} = 1.903$$

Procedure:

* Connect circuit shown in figure 1 on breadboard
* Connect CRO Ch_1 to input and ch_2 to output
* Set the input voltage to $4V_{p-p}$
* Vary the frequence from 500 Hz to 3 KHz in small steps to get a maximum output voltage magnitude. The frequency at the maximum voltage to resonant frequency
* Calculate the bandwidth by subtracting the 2-cutoff frequencies
* Calculate Q factor
* Repeat the steps 1 through 7

Marks split up:

Sl. No	Description	Marks allocated	Marks obtained
1.	Experimental setup	20	20
2.	Observation	20	20
3.	Calculation	20	10
4.	Viva	20	10
5.	Record	20	10

Result:

Thus the phenomenon of resonance in RLC circuit was studied and the following were determined.

23/10/24